

1. Administrative & Study Identification

Full Study Title: A Phase 1/2, Open-Label, Adaptive, Dose-Ranging Study to Assess the Safety and Tolerability of PF-07055480/ Giroctocogene Fitelparvovec Gene Therapy in Adults With Severe Hemophilia A **Short Title/Acronym:** Alta Study **Protocol Number:** C3731001 (also SB-525-1603) **NCT Number:** NCT03061201 **Sponsor Name:** Pfizer Inc. (in partnership with Sangamo Therapeutics)
Investigational Product: Giroctocogene fitelparvovec (PF-07055480 / SB-525), a recombinant AAV6 vector carrying the B-domain-deleted human F8 gene **Phase of Development:** Phase I/II
Medical Monitor: Pfizer Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety, tolerability, and changes in circulating factor VIII (FVIII) activity after a single intravenous dose of giroctocogene fitelparvovec. **Secondary Objectives:** To evaluate the use of exogenous FVIII replacement therapy (Annualized Infusion Rate) and the frequency of bleeding events (Annualized Bleeding Rate).

2.2 Background & Rationale Patients with severe hemophilia A have missing or defective FVIII, requiring frequent exogenous FVIII infusions or nonfactor hemostatic agents to prevent spontaneous and life-threatening bleeding events. Adeno-associated virus (AAV) vector-based gene therapy is designed to deliver a functional copy of the F8 gene directly to liver cells. This allows for endogenous FVIII production, potentially offering a one-time treatment that reduces or eliminates the lifelong need for FVIII replacement therapies.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm, dose-ranging **Blinding:** Open-label **Randomization:** N/A (Sequential cohorts with adaptive dose escalation)

3.2 Planned Enrollment & Duration Target \$N\$: 11 adult male participants **Number of Sites:** ~22 locations (United States)
Estimated Study Start: 21-Jun-2017 **Estimated Primary Completion:** 16-Jul-2024

4. Subject Selection (Eligibility Criteria)

- 4.1 Inclusion Criteria**
1. Male ≥ 18 years of age.
 2. Severe hemophilia A, defined by past evidence of circulating FVIII activity $< 1\%$ of normal.
 3. Previously treated or exposed to FVIII concentrates/cryoprecipitate for ≥ 150 exposure days.
 4. History of ≥ 12 bleeding episodes over the 12 months preceding screening, if on on-demand therapy.
- 4.2 Exclusion Criteria**
1. Presence of pre-existing neutralizing antibodies (NABs) to the AAV6 capsid and/or current/previous history of FVIII inhibitors.
 2. History of Hepatitis B or HIV-1/2 infection; Hepatitis C unless viral assays collected 6 months apart are negative.
 3. Markers of hepatic inflammation, overt/occult cirrhosis, or presence of liver mass.
 4. History of chronic renal disease or creatinine ≥ 1.5 mg/dL.
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5. Intervention & Treatment Plan

- 5.1 Investigational Regimen Dosage/Frequency:** Single intravenous (IV) infusion of giroctocogene fitelparvovec across 4 ascending dose cohorts (9×10^{11} , 2×10^{12} , 1×10^{13} , and 3×10^{13} vg/kg). **Route of Administration:** Intravenous (IV). **Duration of Treatment:** Single-dose administration with an observation and follow-up period lasting 5 years (260 weeks).
- 5.2 Concomitant & Prohibited Therapy Permitted:** FVIII replacement therapy on-demand; systemic corticosteroids (e.g., if alanine aminotransferase [ALT] levels rise $> 1.5 \times$ baseline). **Prohibited:** Use of routine prophylactic FVIII replacement starting ~2 weeks post-infusion (switched to on-demand only).
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6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Anti-AAV6 NAb assay, FVIII inhibitor check
Baseline	Day 0	Admission to infusion center, Single IV administration of SB-525
Post-Infusion	Day 1	24-hour post-infusion observation and discharge
Interim	Weeks 1 to 20	Monitored at least weekly for FVIII activity and twice weekly for hepatic enzymes
End of Study	Week 260 (5 Years)	Long-term efficacy, safety evaluation, final liver imaging, and exit interview

7. Outcome Measures (Endpoints)

- 7.1 Primary Endpoint Definition:** Assessment of safety (incidence of adverse events) and changes from baseline in circulating FVIII

activity levels up to 5 years post-infusion. **Measurement Method:** Chromogenic (CH) assay and One-Stage (OS) clotting assay for FVIII activity.

7.2 Secondary & Exploratory Endpoints Annualized Bleeding Rate (ABR): Frequency of bleeding episodes requiring treatment.
Annualized Infusion Rate (AIR): Use of exogenous FVIII replacement therapy post-infusion.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Intensive monitoring of hepatic enzymes (ALT/AST) for signs of immune response against transduced hepatocytes. Corticosteroids are administered upon enzyme elevation. **Serious Adverse Events (SAEs):** Routine 24-hour reporting. Notable SAEs tracked included acute infusion reactions (e.g., transient hypotension/pyrexia). **Data Monitoring Committee (DMC):** Independent safety monitoring panel evaluating dose-escalation progression.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) / Safety Set for all participants receiving any fraction of the vector infusion.
Sample Size Justification: Adaptive dose-ranging study design with 11 subjects (n=2 in lower dose cohorts, expanded to n=5 in the highest 3×10^{13} vg/kg cohort) deemed sufficient to evaluate preliminary safety and FVIII expression. **Handling of Missing Data:** Descriptive statistics for longitudinal ABR and FVIII activity; missing follow-up data evaluated up to the last known data cutoff.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: Conducted in accordance with Good Clinical Practice (GCP), the Declaration of Helsinki, and the Council for International Organizations of Medical Sciences guidelines.
Monitoring Plan: Strict 24-hour onsite inpatient observation followed by rigorous weekly/bi-weekly lab sampling for 20 weeks.
Audit Trail: Independent review of chromogenic assays and clinical records via centralized electronic data capture (EDC).

11. Study Identifiers & Governance

Primary ID: C3731001 **Registry Number:** NCT03061201 **Sponsor/Lead Organization:** Pfizer Inc. **Study Title:** Alta Study **Official Regulatory Title:** A Phase 1/2, Open-Label, Adaptive, Dose-Ranging Study to Assess the Safety and Tolerability of PF-07055480/

12. Clinical Framework (Hemophilia A Specific)

Primary Condition: Severe Hemophilia A **Diagnosis Threshold:** Circulating FVIII activity $< 1\%$ of normal **Medical Methodology:** Adeno-associated virus (AAV) vector-based gene therapy (AAV6) **Optimization Target:** Sustained endogenous FVIII production sufficient to prevent spontaneous bleeding events

13. Study Procedures & Methodology

Management Pathway: A single intravenous infusion of gene therapy vector with transitioning of patients from prophylactic to on-demand exogenous FVIII replacement. **Education Protocol (HCP):** Strict protocols for evaluating transaminase elevations and triggering rapid corticosteroid intervention. **Education Protocol (Patient):** Guidance on identifying bleeds, ceasing routine prophylaxis 2 weeks post-infusion, and managing on-demand treatments. **Quality Audit Mechanism:** Real-time tracking of FVIII assay discrepancy (chromogenic vs. one-stage) and longitudinal neutralizing antibody development.

14. Observation & Follow-Up Schedule

Total Duration: 260 Follow-up weeks (5 Years) **Onsite Visit Cadence:** 24-hour observation post-dosing, weekly (Weeks 1-20), then scaled to quarterly/annual monitoring. **Remote Monitoring Frequency:** Routine logging of potential bleeding events and exogenous FVIII usage. **Checklist Review Interval:** Routine clinical assessments matching established milestones (e.g., Week 52, 104, 156, 260).

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				